Aryaman Nasare

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Education

University of Illinois Urbana-Champaign — Champaign, Illinois
Master of Computer Science | ***GPA: 3.96 / 4.00*** — *Aug 2024 – Dec 2025*
Coursework *Cloud Computing Applications, Deep Learning for Computer Vision, Software Engineering*

Savitribai Phule Pune University — Pune, India
Bachelor of Computer Engineering, Honors in Data Science | ***GPA: 8.74 / 10.00*** — *Aug 2019 – May 2023*

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, Bash
Frameworks & Libraries: React, Node.js, Express.js, FastAPI, PyTorch, TensorFlow, LangChain, Selenium
Cloud & DevOps: Docker, AWS (Lambda, S3, EC2, IAM), GCP, Git, GitHub Actions (CI/CD), Postman, Linux/Unix
Databases & AI: MySQL, MongoDB, Neo4j, FAISS, ChromaDB, RAG, LLMs, Diffusion Models

Experience

University of Illinois Urbana-Champaign — Champaign, Illinois
Graduate Teaching Assistant – Database Systems & Foundations of Data Curation — *May 2025 – Dec 2025*

- Developed data systems assignments covering SQL schema design and indexing, **NoSQL/graph** systems (MongoDB, Neo4j), and data curation workflows with **Docker**; supported them at scale for **550+ students** across **two** graduate courses.
- Engineered a **Bash** automation script interfacing with the **GitHub REST API** to parse user CSVs and batch-provision organization access, eliminating manual onboarding overhead and saving **15+ hours**.

RIA Advisory — Miami, Florida (Remote)
Software Engineer Intern – Technology Track — *May 2025 – Aug 2025*

- Developed **60+** end-to-end automation flows using **Selenium** to validate the company's flagship **SaaS Test Automation Platform**, implementing dynamic waits and strict assertions to minimize flakiness and expand regression coverage.
- Engineered a **LangChain RAG** pipeline indexing **200+ pages** of internal technical guides into a **FAISS** vector store, empowering **20+ interns** to self-resolve environment issues and reduce troubleshooting time by **33%**.

Reliance Jio Platforms Ltd. — Navi Mumbai, India
Software Engineer Trainee — *Jan 2024 – Jul 2024*

- Orchestrated regression strategies for the JioPartnerWorld application (**1M+ users**), validating cloud-hosted microservices via **Jenkins CI/CD** pipelines and designing end-to-end test scenarios tracked in **Jira**, reducing production defect leakage by **15%**.
- Engineered a **Selenium** automation suite executed via **AWS** CodeBuild to reduce manual workflows, leveraging parallelized cloud execution to cut test execution time by **~90%** (saving **6+ minutes** per test cycle).
- Led **root cause analysis** for workflow faults across distributed systems by analyzing **AWS** CloudWatch logs and **SAP** IDoc failures, collaborating with backend teams to resolve errors and improve **MTTR** by **40%**.

Projects

InsightHub: AI-Powered Team Collaboration Platform | *React, TypeScript, Node.js, Express, MongoDB* — GitHub

- Designed a full-stack **multi-tenant platform** supporting secure document management, team activity notifications and AI-assisted chat, backed by a RESTful Node.js/Express API and MongoDB backend.
- Implemented **JWT auth** using a centralized **Express middleware** guard, enforcing team-membership/creator authorization and CORS allowlisting to ensure strict tenant isolation.
- Developed an **LLM-powered RAG** semantic search and chat pipeline over uploaded documents using **embeddings + vector similarity**, enabling context-aware answers within private team workspaces.

GhibliDream: Few-Shot Image Generation via Diffusion Fine-tuning | *PyTorch, Stable Diffusion, Docker* — GitHub

- Built an end-to-end few-shot DreamBooth fine-tuning pipeline for **Stable Diffusion 2.0**, enabling Studio-Ghibli-style image generation with only **5–7 image–text** pairs; containerized the workflow with **Docker** to enable reproducible reruns.
- Automated LLM-assisted caption rewriting across an **800+ image** dataset and applied a **two-token identifier** fine-tuning strategy to improve subject/background disentanglement, achieving a **0.914** foreground CLIP similarity for subjects.

SWE-Agent: Autonomous Code Repair System | *Python, Docker, LLMs, Bash* — GitHub

- Extended the Mini-SWE-Agent code repair framework to resolve GitHub issues by developing **10 structured tools** (regex search/replace, linter, and memory search) enabling LLM-driven code navigation and patching via a **tool-call protocol**.
- Achieved a **70% resolution rate (28/40)** on **SWE-bench Verified** subset (containing issues from Django, Matplotlib), surpassing the baseline performance (12/40) by **2.3x** through iterative validation inside a Docker sandbox.